



EXPERIMENT NO 9

NO-LOAD AND BLOCKED-ROTOR TESTS ON A 3-PHASE SLIP RING INDUCTION MOTOR

AIM

- To conduct no load and blocked rotor tests on 3 phase slip ring induction motor
- To determine the equivalent circuit parameters and hence predetermine the performance at full load from the equivalent circuit
- To draw the circle diagram and hence predetermine the performance at full load from circle diagram.

APPARATUS

Sl. No.	Name of the apparatus	Type	Range	Quantity
1.	Voltmeter			
2.				
3.				
4.	Ammeter			
5.				
6.				
7.	Wattmeter			
8.				
9.	Rheostat			

PRINCIPLE

Depending upon the construction of rotor, there are two types of 3-phase induction motors - a) squirrel cage and b) wound rotor or slip ring type. A squirrel cage rotor has a number of conducting bars (made of copper or **aluminum**) laid in the slots of the rotor core. These bars are short-circuited at **both** ends by conducting end rings. The cage winding is adaptable to any number of poles. The cage rotor motor is cheap and robust. However its starting torque is low.

A wound rotor has a laminated core with slots on its outer surface. These slots carry 3-phase rotor winding, which is similar to the stator winding. Both the stator and rotor windings are designed for the same number of poles. The 3-phase rotor winding is usually star connected. The ends of three phases are tied to slip rings mounted on the motor shaft. The rotor windings are shorted through brushes, which ride on the slip rings. Thus the rotor currents are accessible at these brushes. Extra resistance can be connected to the slip rings. This extra resistance is usually necessary to give a high starting torque.